

TAXONOMY OF AGRICULTURAL TIME-SERIES FORECASTING APPROACHES

Classification of Modeling Paradigms, Analytical Properties, and the Emergence of Decision Intelligence

Agricultural Price Forecasting Framework

1. Statistical Models

Subcategories:

- Moving Average
- Exponential Smoothing
- ARIMA / SARIMA

Analytical Profile:

Interpretability: High (✓)
Data Requirement: Low (✓)
Adaptability: Low (✗)
Decision Support: None (✗)

2. Machine Learning Models

Subcategories:

- Random Forest / SVR
- Gradient Boosting
- XGBoost / LightGBM

Analytical Profile:

Interpretability: Medium (~)
Data Requirement: Medium (~)
Adaptability: Medium (~)
Decision Support: None (✗)

3. Deep Learning Models

Subcategories:

- LSTM / GRU Networks
- CNN-LSTM Hybrids
- Temporal Transformers

Analytical Profile:

Interpretability: Low (✗)
Data Requirement: High (✗)
Adaptability: Low (✗)
Decision Support: None (✗)

4. Intelligent Decision Systems

Subcategories:

- Ensemble Fusion Architectures
- Explainable AI (XAI) Frameworks
- Volatility-Aware Risk Models
- Multi-Agent Market Intelligence

Analytical Profile:

Interpretability: High (✓)
Data Requirement: Medium (~)
Adaptability: High (✓)
Decision Support: Actionable (✓)

Ensemble Learning

Stacking
Blending
Meta-Learning

Explainable AI (XAI)

SHAP Analysis
LIME Explainer
Rule Engines

Volatility Intelligence

GARCH / EGARCH
Markov States
Regime HMM

Market Intelligence

Granger Causality
VAR Modeling
Spillover Net

MANDISENSE AI (PROPOSED ARCHITECTURE)

- Unified Multi-Agent Decision Framework
- Specialized Agents: Seasonality, Arrival, External Intelligence
 - Adaptive Blending: Alpha Ridge Meta-Ensemble Fusion
 - Risk Integration: GARCH-HMM Volatility & VAR Spillovers
 - Cognitive Delivery: LLM-Driven Advisory & Actionable Signals

TAXONOMY KEY & METRICS

✓ : Highly Optimized / Capable
~ : Partially Optimized
✗ : Limited / Not Present

Note: Traditional approaches focus strictly on numerical forecasting, lacking contextual intelligence.

CORE TAXONOMICAL INSIGHT: Forecasting methodologies evolve from rigid, prediction-focused mathematical models toward adaptive, explainable, and decision-oriented multi-agent intelligence systems.